



## AQ9.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Functions of several variables:

#### Level 1:

1 point

Which of the following are correct with respect to  $f(x, y) = x^2 - 2y$ , where  $x, y$  are real?

☐

Domain of  $f$  is  $\mathbb{R}^2$

☐

Domain of  $f$  is  $\mathbb{R}$

☐

Range of  $f$  is  $\mathbb{R}$

☐

Range of  $f$  is  $\mathbb{R}^2$

**No, the answer is incorrect.**

**Score: 0**

#### Accepted Answers:

Domain of  $f$  is  $\mathbb{R}^2$

Range of  $f$  is  $\mathbb{R}$

1 point

Let  $D \subset \mathbb{R}^n$  and  $f: D \rightarrow \mathbb{R}^m$ ,  $g: D \rightarrow \mathbb{R}^m$  be multivariable functions on  $D$ . Which of the following statements is(are) TRUE?

☐

The product function  $f \cdot g$  is defined on  $D$  by  $(f \cdot g)(x) = f(x) \times g(x)$ ,  $x \in D$

☐

The sum function  $f + g$  is defined on  $D$  by  $(f + g)(x) = f(x) + g(x)$ ,  $x \in D$

☐

Let  $c \in \mathbb{R}$ . The function  $cf$  is defined on  $D$  by  $(cf)(x) = c \times f(x)$ ,  $x \in D$

☐

If  $m = 1$  and  $g(x) \neq 0$ ,  $x \in D$ , then the function  $f/g$  is defined on  $D$  by  $(f/g)(x) = f(x)/g(x)$ ,  $x \in D$

**No, the answer is incorrect.**

**Score: 0**

#### Accepted Answers:

The sum function  $f + g$  is defined on  $D$  by  $(f + g)(x) = f(x) + g(x)$ ,  $x \in D$



If  $m = 1$  and  $g(x) \neq 0$ ,  $x \in D$ , then the function  $f/g$  is defined on  $D$  by  $(f/g)(x) = f(x)/g(x)$ ,  $x \in D$ .

1 point

Suppose  $f: D \rightarrow \mathbb{R}$  is a function defined on a domain  $D \subset \mathbb{R}$ . Which of the following can be  $f$ ?

☐

$$f(x) = 4 \pm \sqrt{x-10} \quad f(x) = 4 \pm x - 10$$

☐

$$f(x) = \frac{x^2}{x+1} \quad f(x) = x+1x^2$$

☐

$$f(x) = \log(\sqrt{x^4 + 3}) \quad f(x) = \log(x^4 + 3)$$

☐

$$f(x) = x^3 + x^2 - x \quad f(x) = x^3 + x^2 - x$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f(x) = \frac{x^2}{x+1} \quad f(x) = x+1x^2$$

$$f(x) = \log(\sqrt{x^4 + 3}) \quad f(x) = \log(x^4 + 3)$$

☐

$$f(x) = x^3 + x^2 - x \quad f(x) = x^3 + x^2 - x$$

1 point

Which of the following statements is(are) TRUE?

☐

The output of a scalar valued multivariable function is always a real number.

☐

The output of a vector valued multivariable function can be a real number.

☐

The output of a single variable vector valued function is always a vector.

☐

The output of a vector valued multivariable function can not be a real number.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The output of a scalar valued multivariable function is always a real number.

The output of a single variable vector valued function is always a vector.

The output of a vector valued multivariable function can not be a real number.

1 point

Consider the function  $S(t) = (\cos t, \sin t, 2t)$ , where  $t$  is real. Choose the set of correct options.

☐

$S$  is a single-variable vector-valued function

☐

$S$  is a multi-variable vector-valued function

☐

The domain of  $S$  is  $\mathbb{R}$  and the co-domain of  $S$  is  $\mathbb{R}^3$

☐

The domain of  $S$  is  $\mathbb{R}^3$  and the co-domain of  $S$  is  $\mathbb{R}^3$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$SS$  is a single-variable vector-valued function

The domain of  $SS$  is  $\mathbb{R}$  and the co-domain of  $SS$  is  $\mathbb{R}^3$

How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi} e^{x^2 + y^2}$  •  $f(x, y) = 3\pi 1 e^{x^2 + y^2}$
- $f(x, y) = (x^3, y^2)$  •  $f(x, y) = (x^3, y^2)$
- $f(x, y) = (x + y, x - y, 2xy)$  •  $f(x, y) = (x + y, x - y, 2xy)$
- $f(x, y, z) = x^2 + xyz - z^2$  •  $f(x, y, z) = x^2 + xyz - z^2$
- $f(x) = (x + 1, x - 1)$  •  $f(x) = (x + 1, x - 1)$
- $f(x, y, z) = (e^{x+y+z}, 0, 0)$  •  $f(x, y, z) = (e^{x+y+z}, 0, 0)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 3

1 point

1 point

Suppose  $g: D \rightarrow \mathbb{R}^3$   $g: D \rightarrow \mathbb{R}^3$  is a function defined on a domain  $D \subset \mathbb{R}^2$   $D \subset \mathbb{R}^2$ . Which of the following functions may be  $g$ ?

☐

$$g(x, y) = 4x + 3y - 12xy \quad g(x, y) = 4x + 3y - 12xy$$

☐

$$g(x, y) = (x^4 + 1, y^4 + 1, xy) \quad g(x, y) = (x^4 + 1, y^4 + 1, xy)$$

☐

$$g(x, y, z) = (e^x - e^z, e^z - e^y) \quad g(x, y, z) = (e^x - e^z, e^z - e^y)$$

☐

$$g(x, y) = (\sin(x + y), \cos(x - y)) \quad g(x, y) = (\sin(x + y), \cos(x - y))$$

☐

$$g(x, y) = (0, \sin(e^x), \cos(e^x)) \quad g(x, y) = (0, \sin(e^x), \cos(e^x))$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$g(x, y) = (x^4 + 1, y^4 + 1, xy) \quad g(x, y) = (x^4 + 1, y^4 + 1, xy)$$

$$g(x, y) = (0, \sin(e^x), \cos(e^x)) \quad g(x, y) = (0, \sin(e^x), \cos(e^x))$$

**Level 2:**

1 point

Which of the following are correct with respect to  $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$   $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$

☐

, where  $x, y$  are real?

☐

Domain of  $f$  is  $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$   $D = \{(x, y) \mid 9x^2 + 16y^2 \leq 1\}$

☐

Domain of  $f$  is  $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$   $D = \{(x, y) \mid 9x^2 + 16y^2 \geq 1\}$

☐

Range of  $f$  is  $[0, 1][0, 1]$

☐

Range of  $f$  is  $[0, 1)[0, 1)$

☐

Range of  $f$  is  $[1, \infty)[1, \infty)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Domain of  $f$  is  $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$   $D = \{(x, y) \mid 9x^2 + 16y^2 \leq 1\}$

Range of  $f$  is  $[0, 1][0, 1]$

1 point

A function  $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$   $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  has the form  $F(x, y) = (P(x, y), Q(x, y), R(x, y))$   $F(x, y) = (P(x, y), Q(x, y), R(x, y))$ . Choose the correct option(s).

☐

$F$  is a multi-variable vector-valued function

☐

$F$  is a single-variable vector-valued function

☐

$P, Q, R, P, Q, R$  are single-variable vector-valued functions

☐

$P, Q, R, P, Q, R$  are multi-variable vector-valued functions

☐

$P, Q, R, P, Q, R$  are multi-variable scalar-valued functions

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$F$  is a multi-variable vector-valued function

$P, Q, R, P, Q, R$  are multi-variable scalar-valued functions

1 point

Consider  $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$   $f(x, y) = \sqrt{1 - 9x^2 - 4y^2}$

☐

. Note: An ellipse is of the form  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$   $a^2x^2 + b^2y^2 = 1$  where  $a, b$  are the dimensions of the ellipse.

☐

Ellipses of increasing size represent the curves  $\{(x, y) \mid f(x, y) = c\}$   $\{(x, y) \mid f(x, y) = c\}$ , where  $c$  increases from 0 towards 1

☐

Ellipses of decreasing size represent the curves  $\{(x, y) \mid f(x, y) = c\}$   $\{(x, y) \mid f(x, y) = c\}$ , where  $c$  increases from 0 towards 1

☐

Ellipses of constant size represent the curves  $\{(x, y) \mid f(x, y) = c\}$   $\{(x, y) \mid f(x, y) = c\}$ , where  $c$  increases from 0 towards 1

☐

The curves  $\{(x, y) \mid f(x, y) = c\}$   $\{(x, y) \mid f(x, y) = c\}$ , where  $c \in [0, 1)$   $c \in [0, 1)$ , cannot be represented by an ellipse

☐

$f(x, y) = 1$  is a point

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Ellipses of decreasing size represent the curves  $\{(x, y) \mid f(x, y) = c\}$ , where  $c$  increases from 0 towards 1

$f(x, y) = 1$  is a point

1 point

Consider a point source  $S = (1, 2, 3)$  radiating energy. The intensity  $I$  at a given point

$P = (x, y, z)$  in space is inversely related to the square of the distance  $d$  between  $S$  and

$P$  :  $I(x, y, z) = \frac{k}{d^2}$ , where  $k$  is a real positive constant. Choose the correct option(s).

☐

Intensity  $I$  is constant on  $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$

☐

Intensity  $I$  is constant on  $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$

☐

Intensity  $I$  is constant on  $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$

☐

Intensity  $I$  is constant on  $\{(x, y, z) \mid x^2 + y^2 + z^2 = 2\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Intensity  $I$  is constant on  $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$

Intensity  $I$  is constant on  $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$